

Reproducibility

Database Description

- * I used the City of Toronto's Daily Shelter & Overnight Service Occupancy & Capacity databased (2025)
- * URL: <https://open.toronto.ca/dataset/daily-shelter-overnight-service-occupancy-capacity/>
- * This data set provides a daily list of active overnight shelter and allied services in the Toronto Shelter and Support Services division's Shelter Management Information System (SMIS) database. The data provides daily updated information about shelter and overnight service programs administered by TSSS including the program's operator, location, classification, occupancy and capacity.

Visualization Purpose

- * I will create visualiztion correlating between Month and the average occupancy rate in ALL shelters in the city of Toronto - which are devided into two type: bed shelters for individuals (coded as "OCCUPANCY_RATE_BEDS") and room shelters for families ("OCCUPANCY_RATE_ROOMS")
- * Month variable does not exit in my dataset, therefore I will create a month variable from ("OCCUPANCY_DATE")
- * Every date includes many observations of all bed/room shelters in Toronto, there for I will have to create a new variable showing the avarage bed occupancy rate per date.

1– Bed and Room occupancy rate are complementary in the original data set;therefore, they were combined into one sheet

2– occupancy rate are the dependent variables; sector is the factoring variable (independent)

3– I grouped occupancy rate by sectors (men, women, families, mixed adults, and youth)

4– I performed one way annova test showing there is a significant difference between groups/ sectors

5 – After calculating standard deiations represented the anova results in two different bar charts including SD and avarage in occupancy rate across sectors

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
Families	5135	501261.68	97.61668549	64.65854861
Men	13102	1263932.11	96.46863914	104.0057153
Mix Adults	14874	1423638.55	95.71322778	160.0825936
Women	9080	888305.6	97.83101322	78.86538148
Youth	9352	874008	93.45680068	173.5023971

ANOVA

Source of Variatio	SS	df	MS	F	P-value	F crit
Between Group	108274.3538	4	27068.58845	217.5064147	1.93E-185	2.37210457
Within Groups	6413883.992	51538	124.4496098			F< Fcrit
Total	6522158.346	51542				
					1.9296826601E-185	P-value <0.05

Groups	Average	Variance	SD
Families	97.61668549	64.65854861	8.041053949
Men	96.46863914	104.0057153	10.19831924
Mix Adults	95.71322778	160.0825936	12.65237502
Women	97.83101322	78.86538148	8.880618305
Youth	93.45680068	173.5023971	13.17203086

Groups	Average	SD
Families	97.67	8.04
Men	96.47	10.19
Mix Adults	95.71	12.65
Women	97.83	8.88
Youth	93.46	13.17

Yearly Occupancy Rate Averages and Standard Deviations (SDs) by Shelter Sector 2025

